# Annex II. Public Debt Sustainability Analysis

*Debt is sustainable under the baseline if the authorities implement their fiscal strategy, but sustainability and financing risks remain significant. General government debt (including guarantees) increased from 36 percent of GDP in 2006 to 78 percent of GDP in 2016. In the baseline, debt is projected to rise to 81 percent of GDP in 2019 before falling to 66 percent of GDP in 2022, based on front-loaded fiscal adjustment measures and the end of highway spending that will improve the fiscal deficit substantially by 2020. The projected debt trajectory is highly susceptible to shocks, especially a negative economic growth shock. Gross financing needs are very high, a particular concern as Montenegro is highly dependent on external financing. Under the baseline scenario, in which the authorities fully implement their fiscal adjustment plan, debt begins a strong downward trend in 2020, but the authorities would still have to manage finances carefully to maintain market access.*

## 1. Baseline and realism of projections

- **Background**. Public debt is reported on a gross basis and includes only general government debt and government-issued guarantees, which comprised 7.9 percent of GDP in 2016.[1] A key driver of the increase in public debt since 2014 has been an external loan to fund construction of the Bar-Boljare highway. Denominated in U.S. dollars, this loan is projected to increase the foreign currency share of public debt to 24 percent by 2020 from 7 percent in 2016.

- **Macroeconomic assumptions**. Growth is projected to increase to 3 percent in 2017, slightly above average growth in 2015–16, as the acceleration of highway construction is partially offset by fiscal adjustment measures.[2] Growth is projected to slow marginally to 2¾ percent in 2018–19 as fiscal adjustment gains steam, but with still-strong highway spending. Growth would then slow in 2020 as the demand stimulus from the highway is withdrawn. Highway expenditures are assumed to add almost 2 percentage points to non-highway GDP over 2015 to 2019, considering a highway-spending multiplier of 0.2 during the construction phase. Over the medium term, growth is expected to reach 3 percent, receiving a modest boost from supply effects related to the completed highway section.

- **Fiscal scenario.** Staff's baseline projections assume a fiscal adjustment of 2 percentage points of GDP in 2017 and the full implementation over 2017–20 of the additional measures outlined in the authorities' fiscal adjustment strategy, which entail and additional fiscal adjustment of 2¼ percent of GDP in the non-highway budget. These adjustment measures

[1] Existing government guarantees are mostly directed to infrastructure and SMEs, often in conjunction with the EBRD. Since peaking at 11.6 percent of GDP in 2012, guarantees fell to 7.9 percent of GDP in 2016. Data on public enterprise debt is not available.

[2] The impact of the highway on growth is explained in more detail in Appendix I and the SIP.

and the path of highway spending explain the surge and fall in the primary deficit over 2015–20. Staff projects that the authorities' fiscal consolidation plan, if fully implemented, will result in a primary surplus of 4½ percent of GDP in 2020, when the primary balance will adjust sharply upward due to the completion of the first phase of the highway.

- **Heat map and debt profile vulnerabilities.** Risks from the debt level are deemed high as debt exceeded the 70 percent of GDP benchmark for the first time in 2014. Debt stays above the benchmark in the baseline projections (except for 2022) and expands substantially under several shock scenarios. Gross financing needs remain above the 15 percent benchmark (except 2018 and 2022), which constitutes a major vulnerability, particularly as Eurobond financing has become more expensive. Public debt held by non-residents (mostly Eurobonds and the Chinese ExIm loan) also constitutes a vulnerability. Growth shocks have a very large impact on the debt profile.

- **Realism of baseline assumptions.** The median forecast errors for real GDP growth and inflation (actual minus projection) in 2007–2015 suggest on average an optimistic bias in staff's past projections, possibly due to the impact of the Great Recession. The median forecast error for the primary balance suggests that staff projections have been in line with outcomes on average. With a large projected adjustment in the cyclically-adjusted primary balance, the key risks are high dependence on external financing and vulnerability to macro shocks. However, the projected adjustment in the primary balance of 8.2 percent of GDP over 2016-20 includes the reduction in highway spending; the primary balance without the highway adjusts by 5½ percentage points.

## 2. Shocks and stress tests

- **Stress tests indicate that growth shocks would have a substantial effect on the debt path.** Fiscal shocks, including a decision to fund the remaining phases of the highway with debt, would also have a major impact on fiscal sustainability.

### *Montenegro-specific stress tests*

- **Highway Phase 2-3.** In this scenario, the authorities decide to proceed with the debt-financed construction of the second and third phase of the Bar-Boljare highway, which staff assumes would cost €1.2 billion, equally spent over 2020–22. Assuming a spending multiplier of 0.2 during the construction phase and modest supply effects upon completion (the same assumptions as for the first phase), economic growth would increase ¾ percentage point on average in 2020–22 relative to the baseline. In this scenario, the primary balance declines in line with highway spending, and the public debt ratio increases to 88 percent of GDP by 2022 (compared to 66 percent in the baseline). Gross financing needs peak at 29 percent of GDP in 2020, 9 percentage points higher than in the baseline. This demonstrates that the authorities cannot afford to take on new debt to complete the second and third phase of the highway over the medium term.

### 3. Standard stress tests

- **Growth shock.** Under this scenario, real output growth rates are lowered during 2018 and 2019 by one standard deviation (4.5 percentage points), also lowering inflation and raising interest rates. The public debt ratio increases peaks at 99 percent of GDP in 2019 before falling to 84 percent of GDP in 2022, while the gross financing needs ratio in 2019 increases 9 percentage points of GDP relative to the baseline.

- **Interest rate shock.** This scenario examines the implications of an increase in interest rates on new debt by 200 basis points (in addition to the baseline assumption of 100 bps) in 2018–22. Debt increases by 3 percentage points in 2022 in this scenario relative to the baseline, while financing needs are affected only marginally.

- **Combined macro shock.** This scenario comprises a recession in 2018 and 2019, a 200 basis-point increase in interest rates, a real exchange rate shock, and a sharp rise in expenditures. It pushes the debt to GDP ratio up by 22 percentage points to 88 percent of GDP by 2022 and substantially impacts gross financing needs.

- **Financial contingent liability shock.** Expenditures increase in 2018 equivalent to 10 percent of the size of the banking sector, combined with a shock to GDP and interest rates. The shock results in a sharp increase in the debt ratio by 17 percentage points to 83 percent of GDP by 2022. Meanwhile, gross financing needs would peak at 27 percent of GDP in 2019 and stay well above the baseline over the medium term.

- **Primary balance shock.** This scenario assumes an expenditure shock and a rise in interest rates leading to a greater than two percentage point deterioration in the primary balance in 2019 and 2020. The combined shocks lead to deterioration in the debt ratio by about 5 percent of GDP by 2022, while the impact on gross financing needs peaks at 3 percentage points of GDP in 2019.

- **Real exchange rate shock.** The scenario assumes a 13 percent depreciation of the real exchange rate in 2018. The Chinese ExIm loan would be issued piecemeal over the course of highway construction. The debt ratio would fall by 2 percent of GDP from the baseline by 2020, based on a larger GDP deflator, with the impact depending critically on the extent of exchange rate pass-through to inflation.

## Montenegro: Public Sector Debt Sustainability Analysis (DSA) - Baseline Scenario

(in percent of GDP unless otherwise indicated)

### Debt, Economic and Market Indicators [1/]

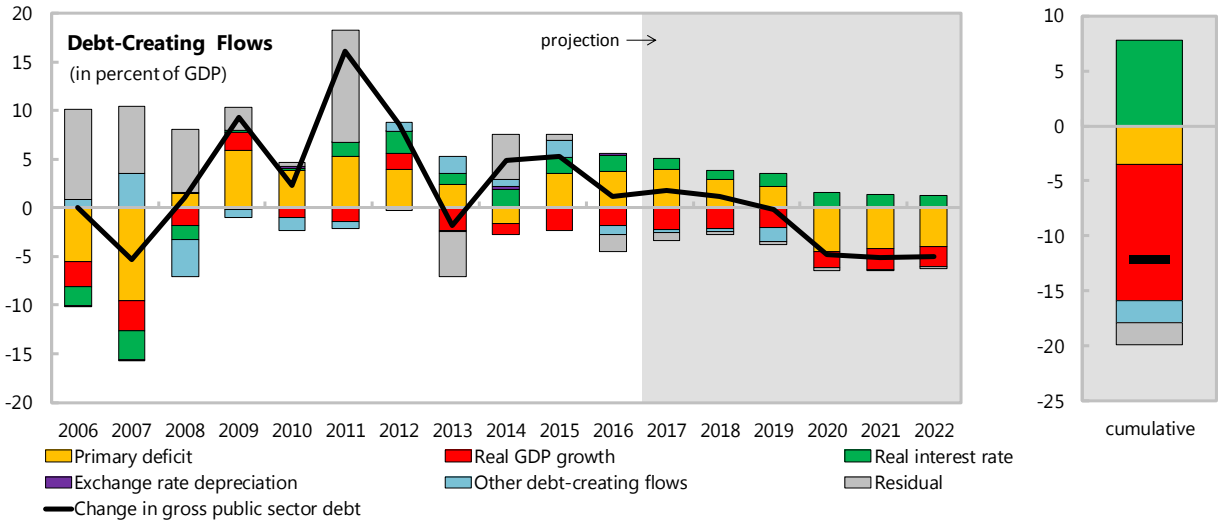
| | Actual | | | Projections | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | 2006-2014 [2/] | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
| Nominal gross public debt | 50.1 | 76.8 | 78.0 | 79.7 | 80.9 | 80.7 | 75.9 | 70.8 | 65.9 |
| Of which: guarantees | 6.6 | 7.5 | 7.9 | 7.7 | 7.3 | 7.0 | 6.7 | 6.4 | 6.1 |
| Public gross financing needs | 4.5 | 14.0 | 21.5 | 16.6 | 14.7 | 22.1 | 20.4 | 16.6 | 11.1 |
| Real GDP growth (in percent) | 3.2 | 3.4 | 2.5 | 3.0 | 2.8 | 2.7 | 2.2 | 3.0 | 3.1 |
| Inflation (GDP deflator, in percent) | 4.3 | 1.4 | 1.6 | 2.2 | 2.6 | 2.0 | 1.7 | 1.8 | 1.9 |
| Nominal GDP growth (in percent) | 7.8 | 4.8 | 4.1 | 5.2 | 5.5 | 4.7 | 4.0 | 4.9 | 5.1 |
| Effective interest rate (in percent) [4/] | 3.6 | 3.9 | 3.4 | 3.8 | 3.9 | 3.9 | 3.9 | 3.8 | 3.9 |

**As of June 26, 2017**

| Sovereign Spreads | | |
|---|---|---|
| EMBIG (bp) 3/ | | 380 |
| 5Y CDS (bp) | | n.a. |
| Ratings | Foreign | Local |
| Moody's | B1 | B1 |
| S&Ps | B+ | B+ |
| Fitch | n.a. | n.a. |

### Contribution to Changes in Public Debt

| | Actual | | | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2006-2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | cumulative | debt-stabilizing primary balance [9/] |
| Change in gross public sector debt | 3.9 | 5.3 | 1.2 | 1.8 | 1.2 | -0.2 | -4.8 | -5.1 | -5.0 | -12.1 | |
| Identified debt-creating flows | -0.2 | 4.7 | 2.9 | 2.6 | 1.5 | 0.2 | -4.6 | -5.0 | -4.8 | -10.1 | -0.8 |
| Primary deficit | 0.7 | 3.5 | 3.8 | 3.9 | 2.9 | 2.2 | -4.4 | -4.2 | -4.0 | -3.5 | |
| Primary (noninterest) revenue and gra | 44.3 | 40.7 | 43.3 | 43.5 | 45.1 | 45.2 | 45.1 | 45.0 | 44.9 | 268.9 | |
| Primary (noninterest) expenditure | 45.0 | 44.2 | 47.1 | 47.5 | 48.1 | 47.4 | 40.7 | 40.9 | 40.9 | 265.3 | |
| Automatic debt dynamics [5/] | -1.0 | -0.6 | 0.0 | -1.0 | -1.2 | -0.6 | -0.1 | -0.8 | -0.8 | -4.6 | |
| Interest rate/growth differential [6/] | -1.0 | -0.6 | -0.2 | -1.0 | -1.2 | -0.6 | -0.1 | -0.8 | -0.8 | -4.6 | |
| Of which: real interest rate | 0.1 | 1.7 | 1.7 | 1.2 | 0.9 | 1.4 | 1.6 | 1.4 | 1.3 | 7.8 | |
| Of which: real GDP growth | -1.1 | -2.3 | -1.8 | -2.2 | -2.1 | -2.1 | -1.7 | -2.2 | -2.1 | -12.3 | |
| Exchange rate depreciation [7/] | 0.0 | 0.0 | 0.2 | ... | ... | ... | ... | ... | ... | ... | |
| Other identified debt-creating flows | 0.1 | 1.8 | -0.9 | -0.3 | -0.3 | -1.4 | 0.0 | 0.0 | 0.0 | -2.0 | |
| Privatization and deposits (negative) | -0.5 | 1.8 | -0.9 | -0.3 | -0.3 | 0.0 | 0.0 | 0.0 | 0.0 | -0.7 | |
| Contingent liabilities | 0.6 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | |
| Use of foreign account receivable | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | -1.3 | 0.0 | 0.0 | 0.0 | -1.3 | |
| Residual, including asset changes [8/] | 4.1 | 0.6 | -1.7 | -0.8 | -0.3 | -0.3 | -0.2 | -0.2 | -0.2 | -2.0 | |




Source: IMF staff.

1/ Public sector is defined as general government and includes public guarantees, defined as Public enterprises.

2/ Based on available data.

3/ Long-term bond spread over German bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes changes in the stock of guarantees, asset changes, and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

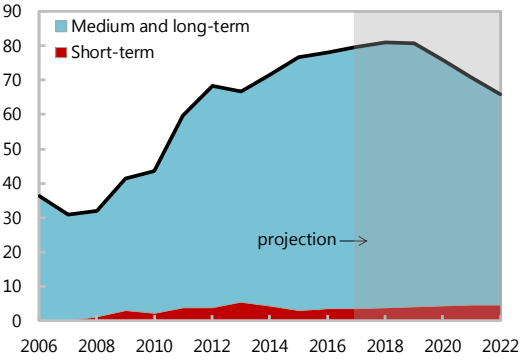
9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

## Montenegro: Public DSA - Composition of Public Debt and Alternative Scenarios

### Composition of Public Debt

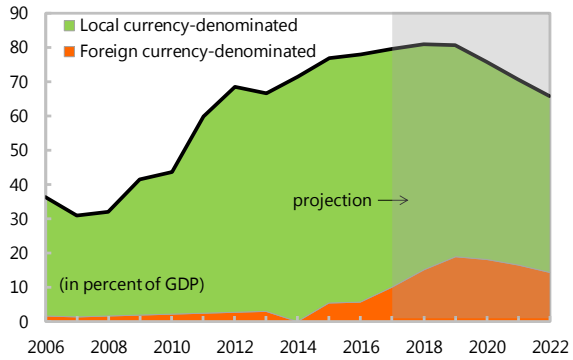
**By Maturity**

(in percent of GDP)

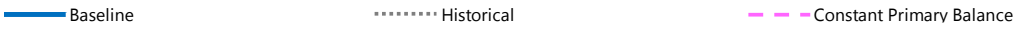



**By Currency**

(in percent of GDP)

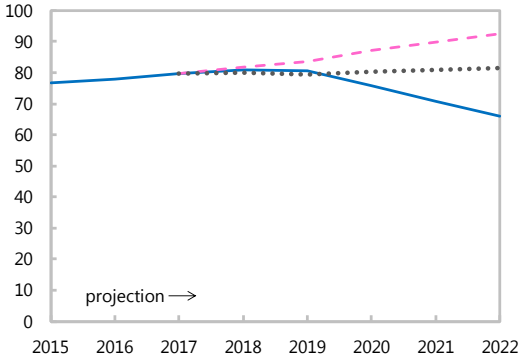



### Alternative Scenarios



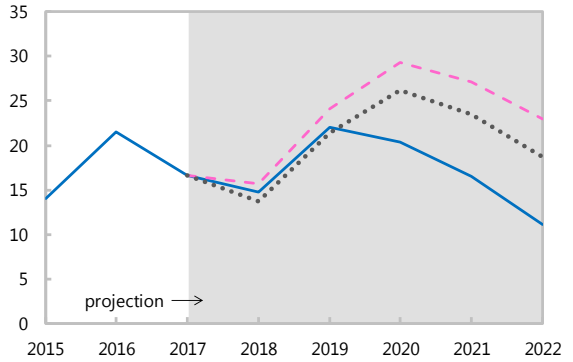

**Gross Nominal Public Debt**

(in percent of GDP)




**Public Gross Financing Needs**

(in percent of GDP)




### Underlying Assumptions

(in percent)

| Baseline Scenario | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
|---|---|---|---|---|---|---|
| Real GDP growth | 3.0 | 2.8 | 2.7 | 2.2 | 3.0 | 3.1 |
| Inflation | 2.2 | 2.6 | 2.0 | 1.7 | 1.8 | 1.9 |
| Primary Balance | -3.9 | -2.9 | -2.2 | 4.4 | 4.2 | 4.0 |
| Effective interest rate | 3.8 | 3.9 | 3.9 | 3.9 | 3.8 | 3.9 |
| **Constant Primary Balance Scenario** | | | | | | |
| Real GDP growth | 3.0 | 2.8 | 2.7 | 2.2 | 3.0 | 3.1 |
| Inflation | 2.2 | 2.6 | 2.0 | 1.7 | 1.8 | 1.9 |
| Primary Balance | -3.9 | -3.9 | -3.9 | -3.9 | -3.9 | -3.9 |
| Effective interest rate | 3.8 | 3.9 | 3.9 | 3.8 | 3.8 | 3.9 |

| Historical Scenario | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
|---|---|---|---|---|---|---|
| Real GDP growth | 3.0 | 2.6 | 2.6 | 2.6 | 2.6 | 2.6 |
| Inflation | 2.2 | 2.6 | 2.0 | 1.7 | 1.8 | 1.9 |
| Primary Balance | -3.9 | -1.9 | -1.9 | -1.9 | -1.9 | -1.9 |
| Effective interest rate | 3.8 | 3.9 | 3.7 | 3.4 | 3.0 | 2.8 |

Source: IMF staff.